Dang Trung Hieu

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About me: Environmental consultant combining biodiversity fieldwork with data engineering and AI. Builds field data systems, automated survey pipelines and ecological AI workflows (camera trap ML, LLM tools) for EIA/EMMP projects in Singapore, with stakeholder experience across NParks, NEA and LTA.

EDUCATION

Nanyang Technological University — **Singapore**

Bachelor's Degree – Environmental Engineering – Honours Merit — Aug 2021 – May 2025

- MH2814 – Probability and Statistics: A+
- Final Year Project: ML model development and validation to predict membrane fouling in water treatment

EXPERIENCE

ESC Environment Singapore — Hybrid

Associate Consultant – Data, Software & AI Solutions — Feb 2026 – Present

- Built a company-wide field data mobile app used by all surveyors for all projects, with GPS, timestamped photos, cloud sync and dashboards, eliminating manual data transcription
- Designed a company-wide ecological database for all projects with species and conservation status referencing NParks SRDB3 and IUCN Red List
- Designed standardised data architecture and data standards, improving data quality, traceability and reuse
- Automated Python/R pipelines for analysis and exceedance detection of monitoring data across all EMMP projects
- Supported EIA reporting with biodiversity analysis (species richness and diversity) and noise, air and water quality assessments for technical report sections
- Automated camera trap videos filtering with Addax AI (MegaDetector), removing false-trigger videos, validated at 95% accuracy
- Built an LLM-assisted workflow for drafting EMMP proposals and monthly reports in 10 minutes, with human review, now used for all EMMP proposals and monthly reports
- Supporting the migration of 20+ years of survey data (5TB+) into a centralised geospatial platform
- Leading company-wide AI adoption; ran an AI automation training workshop for 100+ staff across all country offices

Associate Consultant – EMMP Project Manager — Aug 2025 – Feb 2026

- Managed 5 concurrent EMMP projects from survey scheduling, client communication to deliverables
- Conducted biodiversity surveys (flora, fauna, and intertidal) and fieldwork and site inspections.
- Delivered preconstruction workplan, monthly monitoring reports, and winning proposals for EMMP tender
- Led stakeholder communication and presentations with client and agencies including NParks, NEA, LTA and DSTA

Singapore–MIT Alliance for Research and Technology Centre — Hybrid

Machine Learning Engineer Intern — May 2024 – Aug 2024

- Built a computer vision model to downscale (x64) Singapore temperature forecasts, achieving 0.45 MAE vs a 2.0 target
- Processed multi-dimensional meteorological data (NetCDF4)

Data Analyst Intern — Jan 2024 – May 2024

- Preprocessed and analysed 8 months of climate data from environmental sensors with Python.
- Produced the statistical analysis and figures for a paper published in Urban Climate. (**Publication**)
- Secured an extended internship based on strong technical contributions

Scala Center – Google Summer of Code — Remote

Software Developer — May 2023 – Aug 2023

- Implemented a WebSocket API for http4s Ember, a Scala web framework, empowered faster data transport (**Link**)
- Achieved all project deliverables and presented project at the Northeast Scala Symposium

AWARDS

NTU PEAK Leadership: Sustainability – 1st Place (of 11 teams) — Jan 2025 – Mar 2025

- Partnered with SATS to reduce single-use plastic in large-scale operations; analysed sustainability report data and conducted on-site assessments to identify process gaps
- Proposed a 3-step solution projected to triple plastic recycling and halve plastic use; presented to SATS stakeholders and industry judges

PROJECTS

- **bank.green**: Open-source contribution to a site rating banks on environmental harm
- **SquerryDB**: Full-stack app (PostgreSQL, GraphQL, FastAPI, Next.js) querying 11M IMDB movies/series records (**Link**)
- **Blog**: Analysis of 6 years of personal music streaming with interactive visualisations (**Link**)

SKILLS

- **Programming Languages**:Python, R, SQL, TypeScript/JavaScript, Scala
- **Software and Tools**: QGIS, GeoPandas, PostgreSQL, FastAPI, Next.js/React, Git, Power BI, Tableau, AddaxAI
- **AI:** Machine learning, computer vision, LLM workflows, AI coding assistants